

Technical customer service

Maintenance and service manual



Structure of the hoist for HPE and DPE machine types

Hydraulic hoist with HKS rotary drive

HKS rotary drive

With an HKS rotary drive, the oil pressure supplied is converted into a rotary movement at the drive shaft. This changes the linear movement of the piston into a rotary movement by means of a multiple opposed coarse pitch on the annular gear, piston and shaft.

Connection P1 – anticlockwise

Connection P2 - clockwise

Maximum operating pressure 210 bar

Maximum pressure set 180 bar – safety setting

Annex 1 – section drawing

Assembly instructions

Clean all parts with compressed air, use acid-free grease to grease all sliding parts and thread lightly.

1. Insert O-rings with bearing rings: no. 2 in sealing ring A, no. 5 in base F and no. 6 in regulating screw G.
2. Fit 2-part piston seal no. 4 (O-ring and sliding ring) as follows:
Insert O-ring, heat up sliding ring in hot water (approx. 90 °C), then push over the piston. If necessary, use a rounded tool to slip it into the sealing groove.
After it has cooled down, the sliding ring returns to its original shape.
3. Fit the 2-part rod seals no. 3 in sealing ring A, no. 8 in regulating screw G.
4. Fit the 2-part rod seals no. 7 in piston C.
5. Insert sealing rings with square interior cross-section: no. 1 in sealing ring A and no. 9 in regulating screw G
6. Push piston C into housing D
7. Push drive shaft B into housing D and into piston C. Use a plastic hammer to assist.
8. Screw sealing ring A into housing D using a face spanner and tighten up.
9. Guide annular gear E into housing D via piston C.
10. Adjust position of key. Piston C must contact the housing stop. This is achieved by rotating drive shaft B to the right (clockwise). Then turn drive shaft B further to the right into the desired position for the key (see drawing no.: 20141301784A).
11. Using the face spanner, turn regulating screw G into base F until the gap between the end of the base and the end of the regulating screw is approx. 8.5 mm. The angle of rotation of the drive is then approx. 136.5°.
12. Screw locknut H onto regulating screw G using the face spanner and tighten up against base F.
13. Push base F into the housing.
Warning: face-end HKS serration on housing D, on both sides of annular gear E and on base F must mesh!
14. Screw in cylinder bolts no. 13 with retaining washers no. 12 and tighten up to 148 Nm.

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Safety devices on the hoist

1. Hydraulic pressure – downward movement 50 bar
2. Fixed throttle (drop speed) for downward movement
3. Fixed end position - downward movement – automatically to 500 mm above floor level,
beyond that only if operator keeps key depressed



4. Trolley lock on basket – trolley fixed in basket and held in place



Adjusting the hoist

- a) mechanically
- b) electrically
- c) hydraulically

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Observe the following points

1. Pressure limit - upwards - 180 bar at pressure-limiting valve
2. Pressure limit - downwards - 50 bar at pressure-limiting valve
3. Speed - approx. 22 sec. – at solenoid valve - raise
4. Top limit point – permanent factory setting
5. Bottom limit point – set in situ – adapted to suit floor
6. On 250 l hopper – short arm – short rod – position of arm
7. On 350 l hopper - long arm – long rod – position of arm
8. Tipping characteristics – depending on length of rod
9. Horizontal basket position
10. Basket stop
11. Trolley lock
12. Position of hoist arm 500 mm above floor

3) Mechanical settings

1. Top limit point – permanent factory setting
on-site modification only in exceptional circumstances
tipping characteristics must be taken into account

2. Bottom limit point – set in situ, as dependent on floor
a) set using the machine feet
b) set at the rotary drive:



1. Remove safety hood – loosen the top 2 slot-headed bolts – tap bolts gently to release the adhesion of the hood seal. Undo bolts completely and remove hood.
2. Undo locknut H.
3. Use face spanner to adjust regulating screw G. The angle of rotation is 136.5° and can be adjusted up to $\pm 10^\circ$. To do this, undo locknut H and set the desired angle of rotation by turning regulating screw G. Tighten locknut H back up against the base (use the face spanner). One rotation of regulating screw G adjusts the angle of rotation by 2.7° - approx. 50 mm.

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Warning: do not unscrew regulating screw G beyond the end of the 60 dia. drive shaft.

Adjust in a clockwise direction further up
Adjust in an anticlockwise direction further down

When adjusting, raise the hoist or lift the hoist arm -this facilitates adjustment at the regulating screw

3. Adjusting tipping characteristics, but also **safe distance** between edge of hopper and hoist basket:

distance between edge of hopper and hoist basket = 25 mm.
adjustment by means of rod length (threaded bushing):
shorten rod distance increased
lengthen rod distance reduced

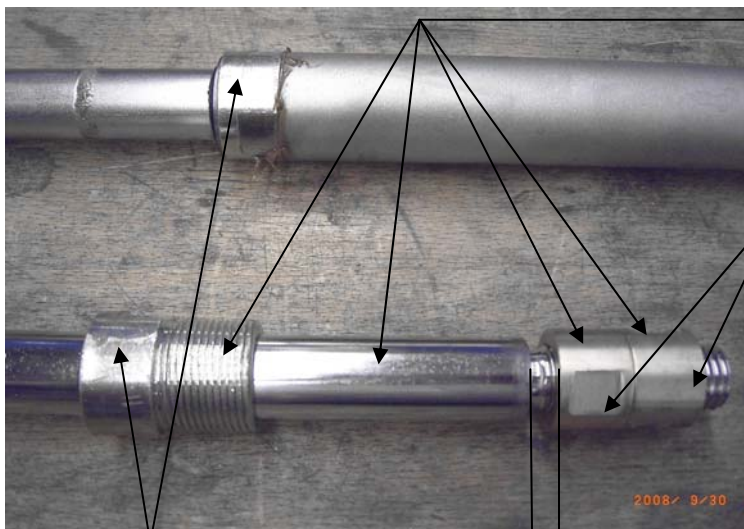
Tilting mechanic from the hoist



External pipe

Inside bar

Tilting arm



Components are bolted in the external pipe

The 2 nuts will be locked to each other. With this you may adjust the measurement **a**.

The more largely the adjusted measured **a** is, the longer becomes the tilting mechanics.

Thus also a later rotation of the wagon takes place. The wagon contents are tilted later into the hopper

Nut with deflector

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4. Modify by means of hopper size

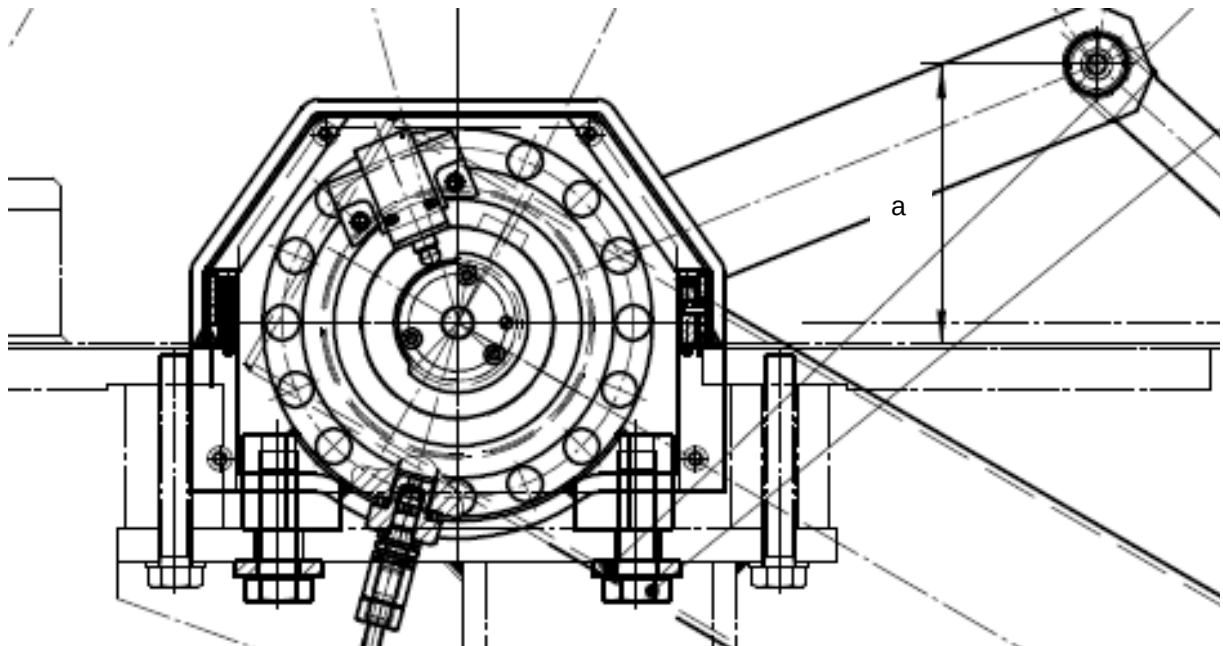
a) Design of hoist arm

250 l hopper short hoist arm – short tipping mechanism
 350 l hopper long hoist arm - long tipping mechanism

b) Position of retaining arm for tipping mechanism

	Size of hopper	HP C / E	DP C / E
Distance – center position connecting eye – machine housing. Measurement a	250 l	144 mm	239 mm
Distance – center position connecting eye – machine housing. Measurement a	350 l	85 mm	180 mm

Drawing with measurement **a**



c) Fitting hoist basket to hoist arm

250 l hopper top screw point (threaded holes)
 350 l hopper bottom screw point (threaded holes)

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Flange to the wagon basket, here the position for 250 liter hopper

Flange to the wagon basket, here the position for the 350 liter hopper.

5. Basket stop

Adjusting the swan-neck rod (bolts on base of machine)
Distance between hoist basket and stop rod 5 mm at bottom limit point.



Distance = Airgap

Is the distance much too big, you will get a swinging wagon basket.

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6. Horizontal position of basket



Adjusting bolt

Counternut

7. Trolley lock



Spring

The spring tension has to be, if not, the trolley protection won't work after driving in the trolley into the basket

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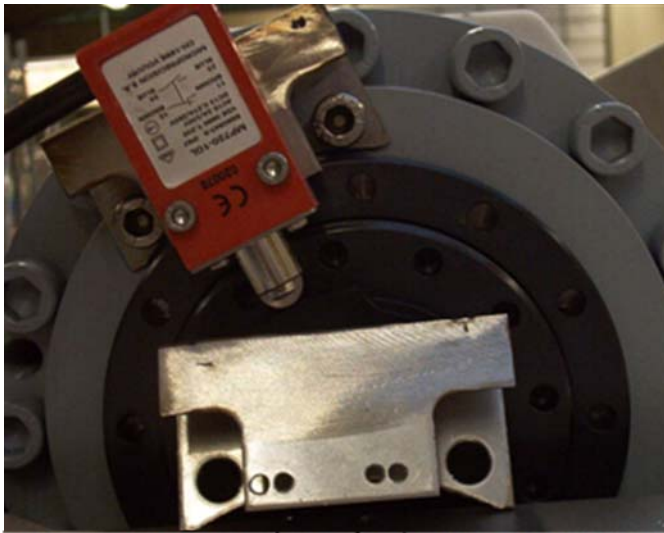


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8) Electrical setting

1. Setting for safety device - hoist automatically lowered to 500 mm above floor - mechanical switch on arm of rotary drive. Lowering can continue only using push-button – accident prevention regulation!
2. Setting 30 sec. at timer relay K3 for valve locking. K3 is on control board A13.



350 l hopper
fixed with screws
in left-hand bores

250 l hopper
fixed with screws
in right-hand bores



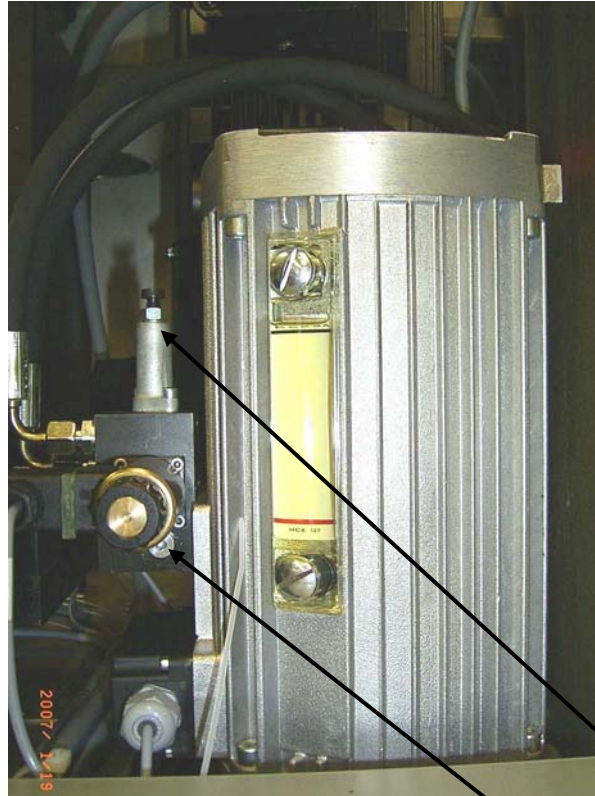
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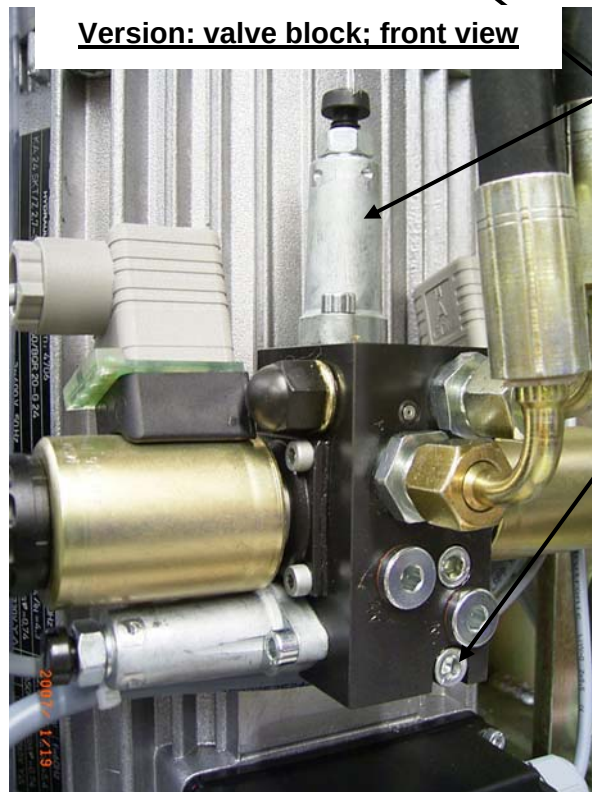


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Version:
Valve block , side view



Version: valve block; front view

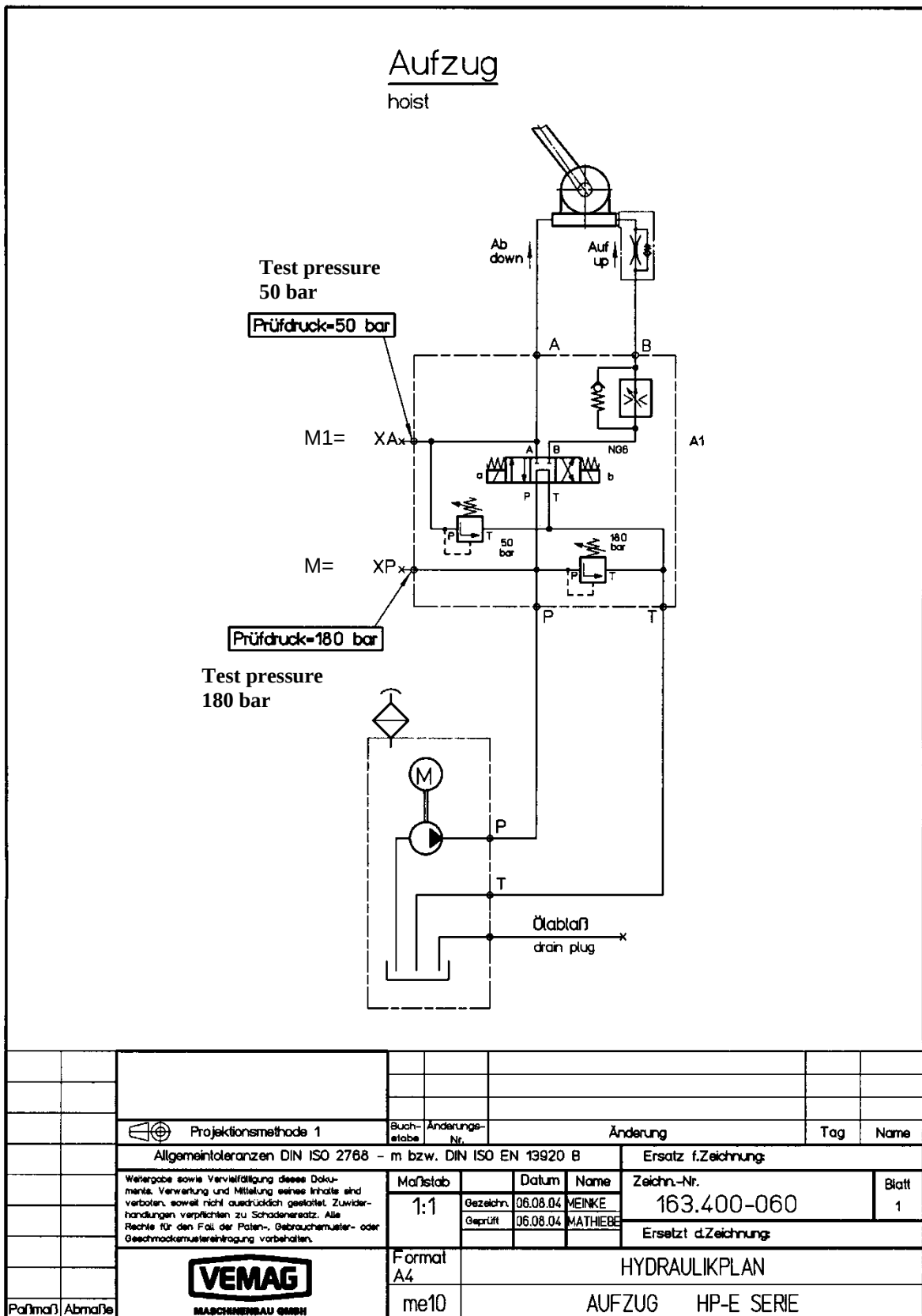
Pressure control valve
Adjustment: operation
pressure 180 bar

Measuring point: **M**
operation pressure
180 bar

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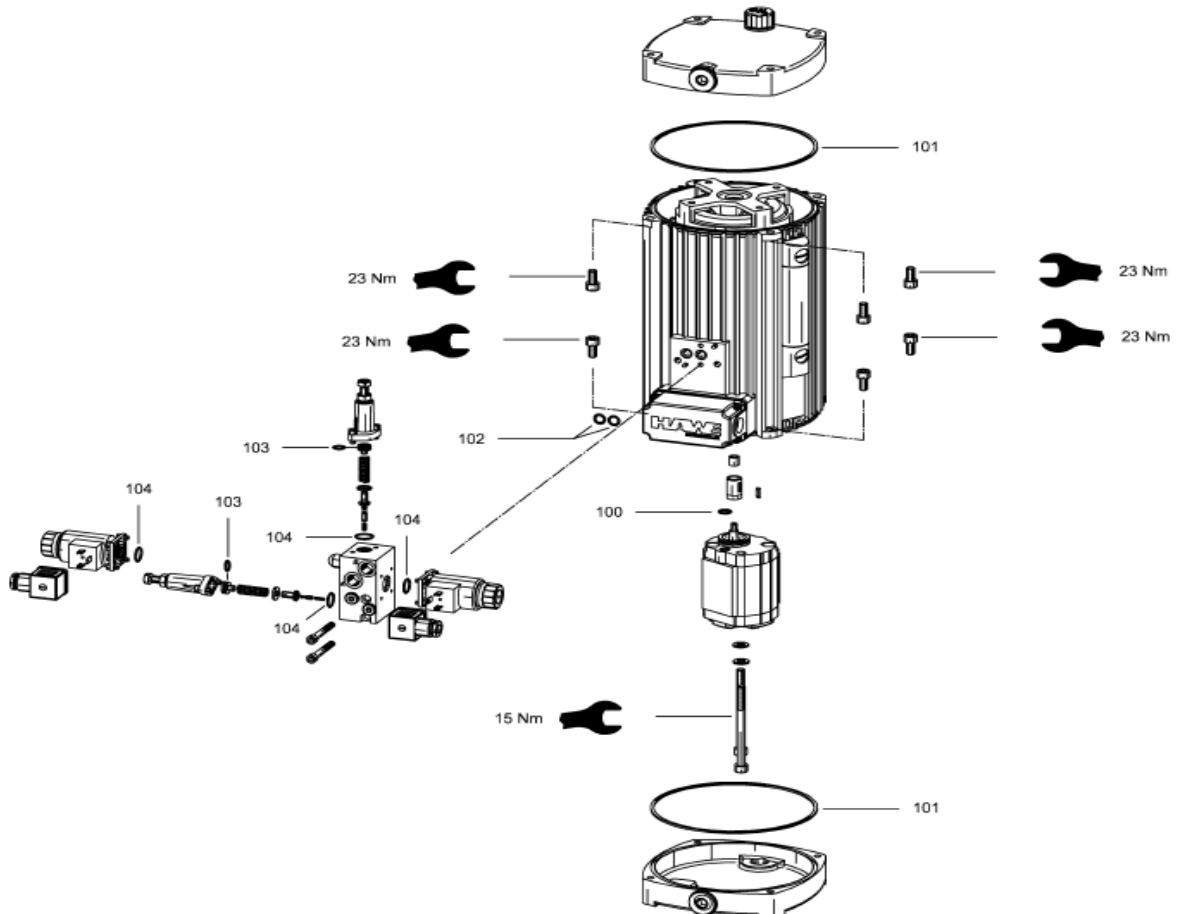
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Exploded drawing from the hydraulic power unit from types:

DS SK 7450 VV

DS SK 7450 VV



Into the seal and gasket set 090.216.304 are included following parts:

HAWE parts number.	Quantity	Position	description	VEMAG number.
6096 400500	1	100	O-Ring 9x2 NBR 90H	090.216.304
6096 981800	2	101	O-Ring 164,77x2,62 NBR90 SH	090.216.304
6096 920800	2	102	O-Ring 8,00x2,00 NBR 90 SH	090.216.304
6096 929200	2	103	O-Ring 9,25x1,78 NBR 70 SH	090.216.304
6096 917100	4	104	O-Ring 15,60x1,78 HNBR 90SH	090.216.304

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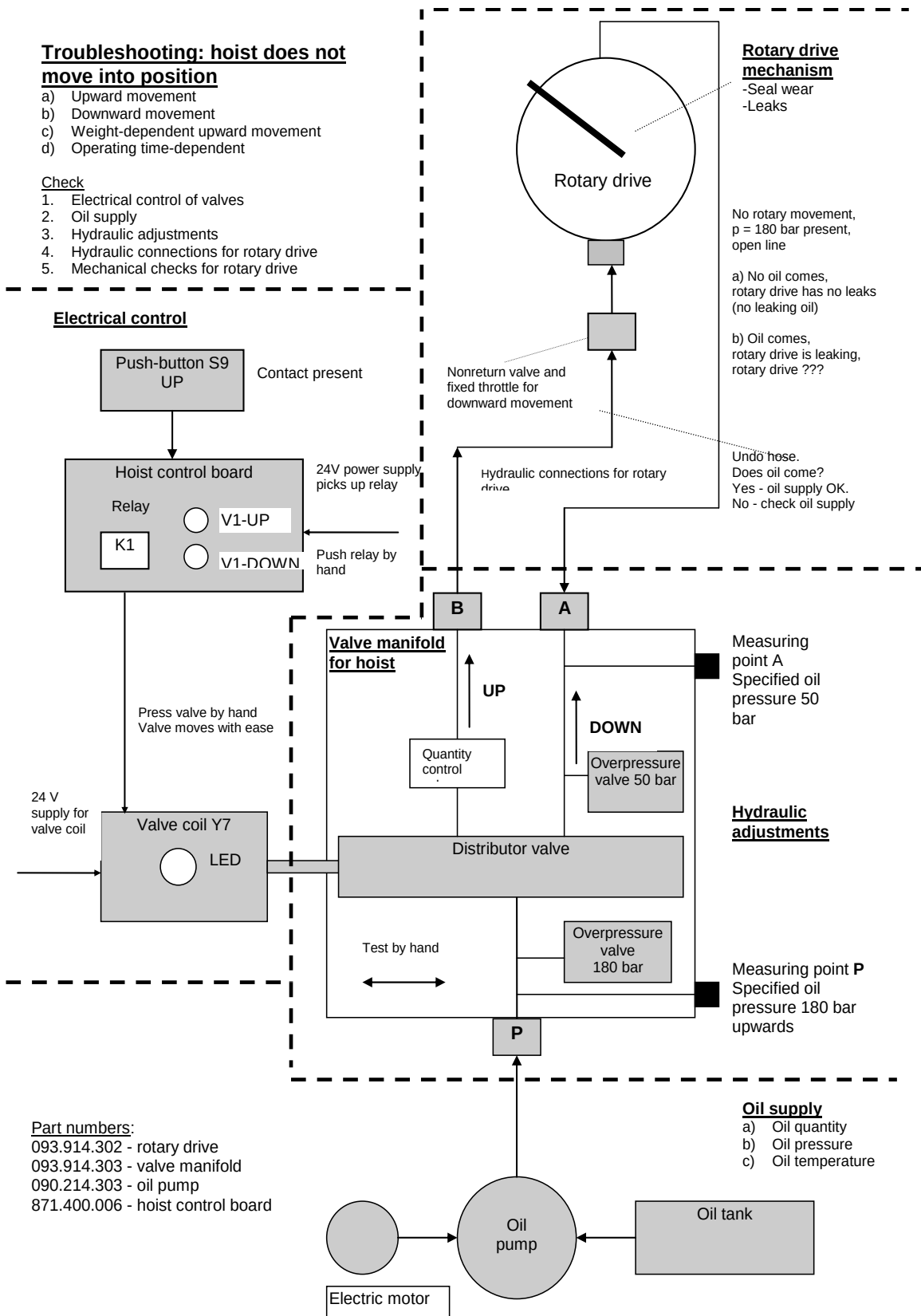
Troubleshooting: hoist does not move into position

- Upward movement
- Downward movement
- Weight-dependent upward movement
- Operating time-dependent

Check

- Electrical control of valves
- Oil supply
- Hydraulic adjustments
- Hydraulic connections for rotary drive
- Mechanical checks for rotary drive

Electrical control



Part numbers:

093.914.302 - rotary drive
 093.914.303 - valve manifold
 090.214.303 - oil pump
 871.400.006 - hoist control board

Oil supply

- Oil quantity
- Oil pressure
- Oil temperature